

A novel vision-based weakly supervised framework for autonomous yield estimation in agricultural applications

Enrico Bellocchio*, Francesco Crocetti, Gabriele Costante, Mario Luca Fravolini, Paolo Valigi

Department of Engineering, University of Perugia, 06125 Perugia, Italy

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ABSTRACT

Autonomous systems have been established as a ground-breaking technology in agriculture, particularly for resource optimization and labor savings. However, even those solutions that are limited to monitoring activities, such as yield estimation, rely on costly robotic platforms equipped with a series of range devices (e.g., LIDAR and GPS-RTK). Recently, vision-based strategies have gained considerable attention as a less expensive and more efficient alternative, capable to be on par with or even surpass approaches that benefit from range sensors. Nonetheless, they exploit deep learning methodologies, which require burdensome labeling procedures to perform training. To address these shortcomings, we present a novel approach that performs yield estimation requiring only a monocular camera and needs a limited amount of supervision information. It detects, locates and maps fruits and tree canopies to estimate the total yield of a specific crop. To keep the image labeling effort to a minimum, we propose a weakly-supervision paradigm that only requires a simple binary label encoding the presence or the absence of fruits in the training images. Our approach does not make any assumptions on the underlying platform, i.e., it can be used by collecting images either with a hand-held camera or with an autonomous robot. Therefore, we are able to considerably reduce the deployment time, the energy and the cost of the overall yield estimation system. At the same time, we keep the performance comparable to both vision-based fully supervised baselines (which require costly labeling operations) and classical systems that rely on more expensive and power-demanding sensors.

1. Introduction

In the last few years, autonomous systems have become increasingly pervasive in agricultural applications. Their use has brought several benefits and allowed the growers to (i) enhance monitoring and control of their cultivation, (ii) optimize the yield operations and ease the maintenance procedures and (iii) save resources and minimize human efforts (Zermas et al., 2020; Sulistyo et al., 2018; Lee et al., 2011; Lambert et al., 2014; Tong et al., 2021; de Araújo et al., 2015). Among the different key operations in agricultural scenarios, yield estimation, whose objective is to measure the total yield of a given crop is certainly one of those that benefited the most from these solutions. Their advantages are particularly evident in contexts with very wide cultivations or in scenarios where human intervention is limited or completely prohibited (e.g., during the current COVID-19 pandemic outbreak).

Most of the State-of-the-Art (SotA) yield estimation systems employ robotic platforms mainly equipped with three types of sensors: range sensors (e.g., LIDAR and sonar sensors), localization sensors (e.g., GPS and GPS-RTK) and vision systems (e.g., monocular, multi-spectral and stereo cameras).

However, each one of those setups is characterized by specific limitations and technical issues. Solutions that rely on LIDAR or differential GPSs are expensive and place important constraints on the overall system, especially with respect to power and weight requirements and the availability of a network of fixed ground reference stations for the GPS correction. These aspects considerably limit the diffusion of such approaches and may not be accessible to small and medium farms with limited financial resources.

For these reasons, the interest in vision-based strategies, that only need low-cost cameras, has rapidly increased in recent years. The impressive results achieved with Deep Neural Networks (DNNs) are currently empowering autonomous systems with high-level reasoning and vision capabilities, making possible the effective application to contexts that only a decade ago were considered to be non-resolvable (Tateno et al., 2017; Ran et al., 2017). Even with just a single camera, it is today possible to detect fruits, track them across frames and localize the camera to avoid multiple counts of the same object and estimate the yield of the entire orchard. Nonetheless, the best performing DNN-based approaches require a huge amount of labeled data (images or videos)

* Corresponding author.

E-mail addresses: enrico.bellocchio@unipg.it (E. Bellocchio), francesco.crocetti@studenti.unipg.it (F. Crocetti), gabriele.costante@unipg.it (G. Costante), mario.fravolini@unipg.it (M.L. Fravolini), paolo.valigi@unipg.it (P. Valigi).

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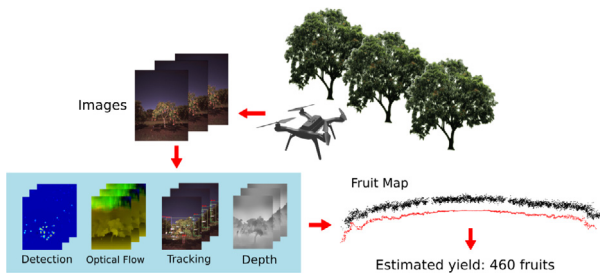


Fig. 1. Overview of the proposed yield estimation system. It exploits a weakly-supervised fruit detector to identify fruits in the images and a trunk/canopy detector to associate them with the trees. Depth and motion estimation modules are used to compute the camera pose and the 3D position of fruits, which are further refined with a graph-based optimization procedure to robustly estimate the yield of the orchard.

in order to perform training. For instance, state-of-the-art strategies for object detection and recognition, which are currently extensively used for yield estimation, are based on Convolutional Neural Networks (CNNs) (e.g., FasterRCNN (Ren et al., 2015), YOLO (Redmon et al., 2016) or MaskRCNN (He et al., 2017)). The optimization of those CNNs needs hundred of thousands of training images and high informative ground truth information (such as bounding boxes and segmentation masks). The acquisition of these labels is costly, burdensome and time-consuming, which considerably slows the deployment phase of the yield estimation system in new crops or orchards. Furthermore, to obtain reliable annotations, expert human operators that are familiar with vision-based detection and segmentation methods need to be involved. If we also consider that it is very hard to train DNNs that robustly generalize over scenarios not observed during training (e.g., unseen fruit categories, different cameras, different weather conditions) without further optimization or fine-tuning procedures, it is clear how the practical applicability of vision-based solutions for yield estimation is further limited.

Inspired by the previous considerations, we propose a novel yield estimation system that benefits from the advantages of vision sensors and, at the same time, exploits a weakly supervised paradigm to heavily reduce the effort needed to train the DNNs (see Fig. 1).

Differently from state-of-the-art methods, our approach is a complete system capable to detect, track and build a graphical map with the position of the detected fruits using only a monocular camera and a minimal amount of supervision.

As we discuss in the following sections, all the modules in our system are designed to keep to a minimum the annotation efforts, by either exploiting weak supervision that requires simple and easy-to-compute labels or by relying on publicly available pre-trained models that do not need further optimization or fine-tuning (e.g., depth and optical flow estimation modules). Furthermore, our approach requires only a monocular camera and does not make any assumption on the underlying platform: the sensor can be either mounted on ground or aerial robots, or hand-held by a human operator. To the best of our knowledge, this is the first attempt to perform vision-based yield estimation exploiting the weakly supervised paradigm.

With different experiments, we demonstrate that our method achieves performances comparable with more burdensome fully-supervised baselines that use monocular cameras or other approaches that rely on more expensive range and localization sensors. Furthermore, we show that it outperforms other weakly supervised systems. Tests are performed on two yield estimation datasets with significantly different setups: the first, containing images from mango orchard, is collected by the Australian Centre of Field Robotics (ACFR) using a ground robot and is presented in Liu et al. (2019), while the second dataset is gathered by the Intelligent Systems, Automation and Robotics Laboratory (ISARLab) using a Micro Aerial Vehicle (MAV) (Bellocchio et al., 2019). These two different setups allow us

to prove the effectiveness of our approach in scenarios with different conditions and challenges.

The paper is organized as follow:

- In Section 2, the related works and the contributions are discussed.
- Section 3 describes the proposed approach.
- Section 4 presents the experiments, with details on the datasets and the baselines used for the comparisons.
- Finally, in Section 5, conclusions are drawn and future works are outlined.

2. Related work

The importance of autonomous systems in precision farming has exponentially grown in the last few years (Thayer et al., 2020; Berenstein and Edan, 2018; Williams et al., 2020; Xiong et al., 2020). For the yield estimation task, alongside range sensors (e.g., LIDAR or lasers), there is increasing adoption of vision-based methods (mostly based on CNNs) that are very effective also in unstructured or partially unstructured environments. A modern vision-based yield estimation approach is a complex system capable to detect, track and map the fruits collecting streams of images. Such a system consists of different functional modules, among which the most relevant is certainly the fruit detection module. Its purpose is to detect and locate the fruits within a single image. In this section, we first provide an overview of the methods used for fruit localization and then we analyze how these methodologies are employed for yield estimation.

2.1. Object detection and counting for agriculture applications

Fully supervised CNN architectures for detection and counting have achieved impressive results in a wide variety of datasets and tasks (Zhang et al., 2015, 2017; Khan et al., 2016; Bargoti and Underwood, 2017a). While their effectiveness is evident and extensively demonstrated, the need for massive amounts of data and supervision labels limits their practical applicability. For these reasons, weakly supervised methods have emerged as a viable and more convenient alternative: their objective is to guarantee performance comparable with fully-supervised baselines while relying on less informative and easier to acquire labels. A diagram depicting the fundamental difference between fully and weakly supervised methods is shown in Fig. 2(a).

In the following paragraphs, we first describe fully-supervised methods used for fruit detection and counting, and then we outline the weakly-supervised approaches.

2.1.1. Fully supervised methods

Fruit detection and counting methods can be divided into detection-based, segmentation-based and regression-based approaches (see Fig. 2(a)). Detection-based strategies rely on CNNs that are optimized to predict the location of fruits in the image. Counting is then achieved by simply enumerating the number of detected fruits. To perform optimization, the training images need to be annotated with ground truth bounding boxes indicating the category and the region of the image containing the object. These strategies have been successfully used in Bargoti and Underwood (2017a), Mai et al. (2020) and Mu et al. (2020). All of them rely on FasterRCNN (Ren et al., 2015) to perform detection. In particular, in Bargoti and Underwood (2017a) the model is fine-tuned to detect apples, almonds and mangoes, while in Mu et al. (2020) the authors focus on tomatoes. Instead, in Mai et al. (2020), FasterRCNN (Ren et al., 2015) is extended to incorporate a multiple classifier fusion strategy and trained to detect oranges. Similarly, in Liu et al. (2020) the authors also consider tomatoes, however, they take advantage of the YOLO (Redmon et al., 2016) architecture.

As an alternative, CNN-based segmentation methods, which perform classification at the pixel level, could be used to measure the number

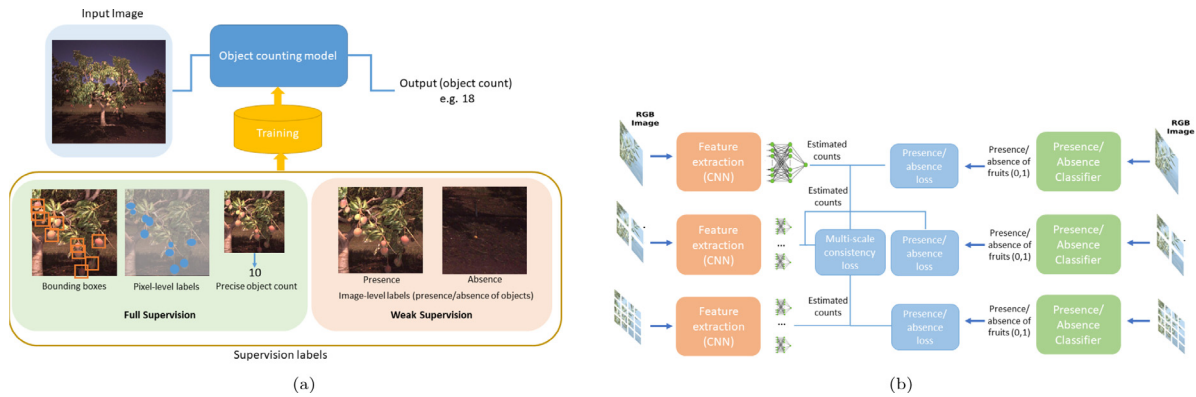


Fig. 2. Characteristics of weakly supervised approaches for fruit counting. Fig. 2(a) depicts the difference between fully and weakly supervised methods. While the former require precise annotations (i.e., bounding boxes, pixel-level labels or object counts), the latter need only presence/absence labels. An example of the optimization procedure for weakly supervised strategies is instead depicted in Fig. 2(b), inspired from Bellocchio et al. (2019). A presence/absence classifier is first trained with image-level labels (weak supervision). Afterward, its predictions are used in combination with the output of the fruit count regression network to impose different consistency losses.

of fruits in images. This idea is exploited in Chen et al. (2017), where the output of a segmentation network is used to detect blobs and count oranges and apples with an auxiliary regression model. More recent segmentation models, such as MaskRCNN (He et al., 2017), are also capable to distinguish different instances of the same class in the image. When trained with images containing fruits, such as strawberries (Yu et al., 2019), sweet peppers (Chen et al., 2020) or apples (Jia et al., 2020), these instances can be used for detection and counting.

The annotation effort can be reduced by using a CNN that regresses directly the fruit count. Specifically, these networks are designed to process an image and output a real number representing the number of fruits. An example of this strategy is presented in Rahnemounfar and Sheppard (2017), where the authors trained a regression network using simulated images to count tomatoes. For comparison purposes with the weakly supervised approach, in Bellocchio et al. (2019) two regression CNNs (single-branched and multi-branched) are presented. Those models are capable to count olives, apples and almonds using the number of fruits as supervision labels.

All the aforementioned approaches address fruit counting by following the fully supervised paradigm. Although their performance is impressive, the collection of the supervision information needed for training is burdensome and time-consuming. This is particularly evident for detection and segmentation-based solutions since they require highly informative labels, i.e., bounding-boxes or pixel-wise annotations. On the other hand, while the labeling effort for regression methods is lower, their output does not contain information to localize the fruits, which is crucial to track them across frames to avoid redundant counts.

2.1.2. Weakly supervised methods

Weakly supervised approaches aim to ease the burden of sample annotation by relying on less informative yet easy-to-acquire labels. Since, in this case, the optimization is not supervised with quantities that explicitly measure the desired output variable, it is common to exploit coherency losses or auxiliary tasks. An example of this process, inspired by the work presented in Bellocchio et al. (2019), is depicted in Fig. 2(b). In this case, weak supervision information (i.e., presence/absence labels) is first used to train a simple presence/absence classifier, whose goal is to predict whether the input image contains fruits or not. Its binary predictions are then combined with the counts regressed at multiple scales to impose different consistency losses and train the object count regression network.

In the literature, there are several weakly supervised approaches used for object recognition, detection, semantic segmentation and instance segmentation (Durand et al., 2017; Zhou et al., 2018; Oquab et al., 2014; Bilen et al., 2015, 2014). Promising works that benefit from the weakly supervised paradigm have also been proposed for

agricultural applications. In Marino et al. (2019) a CNN is trained with image-wise classification labels to locate several types of potato defects within the images. For fruit counting, the approach proposed in Bellocchio et al. (2019), which we summarized in the previous paragraph and in Fig. 2(b), has been extended in Bellocchio et al. (2020), where the authors propose a quasi unsupervised framework that combines a weakly supervised counting network and an unsupervised domain adaptation method. The authors demonstrate that with this strategy it is not necessary to acquire new labels (not even weak ones) when the application domain changes (i.e., the model is used to count a fruit category different from those in the training set).

Despite the interesting results achieved, most of the work previously mentioned are not specifically designed for agricultural tasks. On the other hand, those that address yield estimation (i.e., (Bellocchio et al., 2019) and Bellocchio et al. (2020)) are not designed to locate fruits, thus, there are no straightforward ways to match them across frames and distinguish between new and already tracked elements.

2.2. Yield estimation systems

The works discussed previously focus solely on the detection and counting of fruits in a single image. In this section, we summarize the approaches that integrate the counting module into a complete yield estimation system capable to track and locate fruits. These solutions mostly rely on images acquired by vision sensors mounted on a robotic platform, in some cases with the support of additional sensors such as GPS/GPS-RTK or laser and LIDAR range sensors.

Estimating the yield of an entire orchard raises numerous challenges that cannot be addressed by using only the fruit detection module, in particular:

- **Redundant counts in sequential images:** the same fruit may appear in multiple sequential frames. To avoid redundant counts of the same fruit, it is necessary to distinguish between new and old ones.
- **Occlusions:** Fruits that are visible from a certain point of view might become occluded by foliage or other elements as the camera moves. It is necessary to recognize those fruits once they become visible again and avoid considering them as newly observed.
- **Redundant counts from multiple points of view:** The same fruit can be observed from different points of view (e.g., from two sides of the same cultivation row). Since multiple observations might not be sequential, tracking the fruits across consecutive frames is not sufficient. Instead, fruit positions with respect to the trees and the orchard need to be memorized to recognize them when observed in the future.

Most of these issues can be resolved by using methodologies already adopted for other applications, such as camera pose estimation, localization and mapping. In [Nuske et al. \(2014\)](#) the authors extract hand-crafted features for fruit detection, i.e., Radial Symmetry Transform and Invariant Maximal Detector, from RGB stereo images collected by an Unmanned Ground Vehicle (UGV). These visual cues are used to localize the camera (and, as a consequence, the UGV) and detect fruits in grape orchards. Conversely, the strategy proposed in [Roy and Isler \(2016\)](#) uses a combination of Structure from Motion and SIFT/SURF features to predict apple sizes and perform counting.

Since it is well known that hand-crafted features are not robust to image non-idealities, CNN models, which provide more robust descriptors, have become more popular even in agricultural contexts. In [Stein et al. \(2016\)](#) the authors present a yield estimation framework that combines the information from auxiliary sensors and the visual features extracted by a CNN. The proposed strategy is composed of a fruit detector based on FasterRCNN ([Ren et al., 2015](#)), a LIDAR sensor and a GPS-RTK coupled with an inertial unit. Fruit matching, occlusions and multi-view errors are avoided by estimating the robot pose and by 3D positioning the detected fruits. [Bargoti and Underwood \(2017b\)](#) presents a solution that combines semantic segmentation maps, computed with a CNN architecture and a Multi Layer Perceptron, with GPS data.

In the last years, researchers have focused on strategies that rely only on monocular cameras and remove the need for range and differential positioning sensors. Besides the reduction of the costs, this allows extending the applicability of those systems also in indoor scenarios. Following this direction, [Nellithimaru and Kantor \(2019\)](#) uses only RGB images collected by a UGV for yield estimation within grape orchards. The authors take advantage of MaskRCNN ([He et al., 2017](#)) to perform fruit segmentation. Redundant counts and occlusions are avoided by localizing the robot pose and the fruit locations with a vision-based SLAM method. Multiple-counting errors can be avoided also using Structure-from-Motion, as in [Liu et al. \(2018, 2019\)](#). To perform yield estimation in a mango orchard, the former uses a semantic segmentation CNN while the latter relies on an object detector one. In both cases, occlusions and redundant counts are avoided by combining a Kalman-based object tracker with a semantic Structure-from-Motion algorithm.

More recently, other approaches try also to perform yield estimation with hyperspectral images, such as in [Gutiérrez et al. \(2019\)](#), where this sensor is used to estimate the yield in large mango orchards.

2.3. Contributions

All the weakly supervised approaches described in Section 2.1.2 are designed to regress the number of fruits in a single image. They do not provide strategies to locate and track fruits across consecutive frames. Therefore, they cannot distinguish between fruits already observed in the past or new instances, making it impossible to estimate the yield of a cultivation.

On the other hand, the systems presented in Section 2.2 can track and triangulate into a global 3D map the fruits detected in the images, which allows them to estimate the total yield. However, their fruit detectors have some important drawbacks. Approaches such [Nuske et al. \(2014\)](#) and [Roy and Isler \(2016\)](#) employ hand-crafted features or design heuristic functions that directly process RGB image intensities, which results in systems that are very specific to the fruit species and that lack robustness to non-ideal conditions. Conversely, the strategies proposed in [Stein et al. \(2016\)](#), [Bargoti and Underwood \(2017b\)](#), [Nellithimaru and Kantor \(2019\)](#) and [Liu et al. \(2018, 2019\)](#) exploit the more robust CNN detector. However, they require full supervision to train them and, thus, do not exploit the weakly supervised paradigm. As a consequence, the data collection process for their optimization is burdensome and time-consuming, and limits their practical applicability.

The approach proposed in this work aims to overcome the aforementioned issues. In the following, we summarize our contributions:

- In contrast to the yield estimation systems discussed in Section 2.2 (i.e., [Nuske et al., 2014](#); [Roy and Isler, 2016](#); [Stein et al., 2016](#); [Bargoti and Underwood, 2017b](#); [Nellithimaru and Kantor, 2019](#); [Liu et al., 2018, 2019](#)), we propose a fruit detection approach that benefits from the representational power of CNNs, and, at the same time, exploits the weakly supervised paradigm. In particular, it is designed to require only binary labels encoding the presence or absence of fruits in images for the training phase. These annotations are considerably easier and faster to obtain than bounding boxes or pixel-wise semantic segmentation masks and do not require the intervention of expert operators.
- Our system requires only a monocular camera to operate, avoiding expensive range sensors and the need for complex sensor-rig setups.
- Differently from the weakly supervised approaches presented in [Marino et al. \(2019\)](#) and [Bellocchio et al. \(2019, 2020\)](#), the proposed fruit detection module is integrated into a complete yield estimation system capable to: avoid redundant counts across sequential frames; handle occlusions; estimate the depth of the detected fruits.
- To recognize and avoid multiple counts of the same fruit from different points of view, a 3D global map with fruits and camera poses is incrementally built and refined with a graph-based optimization framework.

3. Proposed approach

In this section, we describe our proposed weakly-supervised yield estimation system. Our framework is composed of several functional blocks (see [Fig. 3](#)):

1. **Fruit detection module**, composed of the Weakly-Supervised Fruit Localization Network (WS-FLNet) described in Sections 3.1.1 and 3.1.2. It is responsible to localize fruits in the image. In particular, it predicts pixel-wise maps indicating the presence and the position of the fruits. To train this module, we do not require pixel-level annotations, instead, we use a binary label that indicates the presence or the absence of fruits within the image. Specifically, we annotate an image with label 0 if it does not contain any fruit and with label 1 when one or more fruits are present, regardless of their number. This module outputs the set of detected fruits described by their 2D coordinates in the image plane.
2. **2D fruit tracking module** (Section 3.1.3). Fruits are tracked on the image plane across frames by combining the detected elements with a Kalman filter-based multi-object tracker. To refine the fruit position estimates during the filter measurement update, we also compute optical flow maps (see FlowNet2 ([Ilg et al., 2017](#)) block in [Fig. 3](#)) encoding the element displacement between consecutive frames. The output of this module consists of a set of tracked fruits (up to the current time step) that are individually described by the 2D pixel position and velocity in the image plane.
3. **Tree canopy/trunk detector and tracking module** (Section 3.2). To associate fruit counts to trees, a specific robust tree detection module is required. This is achieved with a tree canopy/trunk detector. As for the fruits, tree tracking is performed by a Kalman filter. This module provides the image regions containing the tree canopy and trunks tracked up to the current time step.
4. **3D fruit positioning module** (Section 3.3). To avoid the re-counts of the same fruit from multiple points of view, fruits are 3D positioned with respect to the associated tree trunk. The projection of a detected element in the image plane into its 3D world coordinates is computed by estimating its depth using another dedicated functional block implemented by a CNN (see

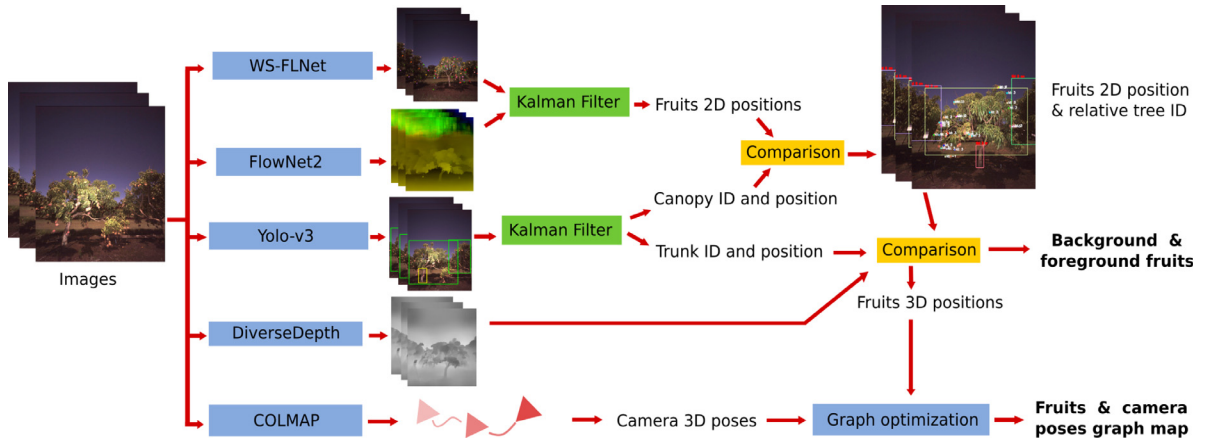


Fig. 3. Overview of the proposed yield estimation framework. Fruits are detected using WS-FLNet and associated across sequential images using a Kalman-based object tracker. The Kalman position estimates are refined by using the optical flow estimated with the FlowNet2 (Ilg et al., 2017) architecture. Similarly, canopies and trunks of the trees are detected with a YOLO-v3 object detector (Redmon and Farhadi, 2018) and tracked by relying on dedicated Kalman filters. The canopy detections are used to associate fruits and trees, while the detected trunks are necessary to distinguish between background and foreground fruits after estimating their 3D positions with the DiverseDepth network (Yin et al., 2020). Camera poses, calculated with COLMAP (Schönberger and Frahm, 2016; Schönberger et al., 2016), are jointly refined with the 3D fruit positions with a graph-based optimizer.

DiverseDepth (Yin et al., 2020) block in Fig. 3). This functionality allows us to distinguish between foreground and background fruits (with respect to the tree trunk) and avoid counting the same element multiple times when observed from the other side of the cultivation row. The module outputs the 3D position of the fruits with respect to the camera reference frame, filtering out those in the background.

5. **Global 3D fruit map estimation module** (Section 3.4). In order to build a map of the orchard with the 3D fruit positions expressed with respect to a fixed global reference frame, we need to estimate the camera trajectory. This is achieved by relying on a Visual Odometry (VO) algorithm. The 3D fruit positions and camera poses are incrementally refined by using a graph optimization procedure. This module produces the global 3D fruit map that can be used to estimate the overall cultivation yield.

It is important to emphasize that the training of the fruit detection module is weakly supervised, while depth maps and optical flows are estimated by using publicly available modules without fine-tuning them. Therefore, the sole annotations needed to optimize these modules are the presence-absence labels of the images. On the other hand, the trunk/canopy detector module is fully supervised with ground-truth bounding boxes. Nonetheless, the time needed to acquire these labels is negligible compared to the annotation effort that a fully supervised fruit detection module would require and, hence, it does not significantly impact the overall deployment cost (quantitative evaluation on the labeling time are provided in Section 4.3). This is even more evident if we consider that trunks and canopies of different fruit species share similar shapes. Therefore, once the trunk/canopy detector is trained with a reasonable variety and amount of samples, it is not necessary to retrain or fine-tune it when the approach is deployed on different agricultural domains.

3.1. Weakly-supervised fruit detection and tracking

The proposed fruit detection module takes inspiration from the work presented in Bellocchio et al. (2019), with the significant difference that in the present study the weak supervision paradigm is exploited to detect and localize fruits, while the aim of the previous study is to simply regress their number. Fruit localization is crucial for yield estimation, since it makes it possible to track and match fruits across frames to avoid redundant counts, and to build a global fruit map, fundamental to analyze the yield distribution in the orchard.

For this purpose, we design a dedicated CNN that outputs high-resolution maps encoding the 2D position of the fruits within the image.

3.1.1. Network architecture

The architecture has an encoder-decoder structure and stems from the popular U-Net (Ronneberger et al., 2015) network (see Fig. 4 and Table 1 for layer details). It is composed of a contractive (encoder) and an expansive (decoder) path, which are symmetric. Both of them are characterized by convolutional, pooling and un-pooling layers. The encoder exploits the residual blocks introduced in the popular Resnet architecture (He et al., 2016) and follows the structure of Resnet50, while the decoder response maps are computed through a series of up-convolutions and concatenations with the high-resolution features extracted by the encoder. This allows adding high-level contextual information to the final prediction, which is crucial to robustly detect fruits, particularly in those cases where occlusions and background-clutter arise. We name this network Weakly-Supervised Fruit Localization Network (WS-FLNet).

More specifically, the first stage of the encoder features a convolutional layer with a 7×7 kernel and 64 feature channels, a batch normalization layer and ReLU (Rectified Linear Unit) activation, followed by a 3×3 max-pooling. The rest of the encoder is built with a sequence of stages with residual blocks. In each residual block, the response maps from the previous one are processed by three convolutional layers (with 1×1 , 3×3 and 1×1 kernels, respectively), each followed by a batch normalization layer and a ReLU activation. As typical in the residual paradigm (He et al., 2016), skip connections are used to add the block input to the feature maps computed by the three convolutional layers. The first stage computes 256 feature maps and is followed by a 3×3 max-pooling layer. The number of filters is doubled after each one of the remaining three stages, resulting in 2048 filter responses at the end of the encoder.

The decoder is based on a sequence of blocks that first up-sample the feature maps and process them with a 3×3 up-convolution halving the number of filters. Afterward, the feature maps are concatenated with the associated responses from the encoder path and then fed into a 3×3 convolutional layer, followed by a final ReLU activation. Each output of the last four blocks is fed into a 1×1 convolutional layer computing 8 response maps at four different scales (i.e., $1/8$, $1/4$, $1/2$ and $1/1$ of the input resolution). After optimization, the 8 maps should have high values in those image regions that contain fruits, thus, we name them *fruit response maps*.

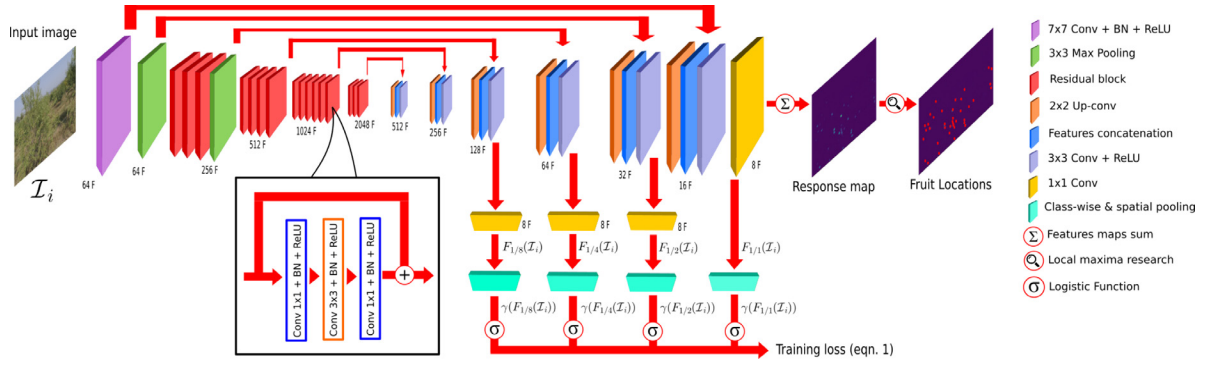


Fig. 4. Overview of the encoder-decoder architecture of the proposed WS-FLNet. The encoding path is characterized by a sequence of residual blocks and follows the structure of Resnet50 (He et al., 2016). The decoder is composed of a sequence of up-convolutions and concatenations with the features extracted by the encoding stages. During training, the output of the last four decoding stages is used to compute the loss function, while, during inference, only the output of the last decoder block is used to estimate the fruit positions.

3.1.2. Weakly supervised network optimization and inference

A possible fully-supervised strategy to train WS-FLNet to estimate fruit locations in the input image would require high-resolution annotated masks. These could be then used to compute a pixel-wise loss between the estimated and the ground-truth fruit maps. However, as mentioned in the above paragraphs, collecting those labels is extremely burdensome, thus, we rely on a weakly-supervised formulation.

In particular, we start by defining a training mini-batch $T_j = \{(I_1, l_1), (I_2, l_2), \dots, (I_M, l_M)\}$ as a collection of M pairs composed by the image I_i and the relative binary fruit presence or absence labels $l_i \in \{0, 1\}$. The objective function is computed by using the fruit response maps at four different scales ($1/8$, $1/4$, $1/2$ and $1/1$ of the input image resolution) from the last up-convolutional blocks in the decoder, i.e., $F_{1/8}(I_i)$, $F_{1/4}(I_i)$, $F_{1/2}(I_i)$ and $F_{1/1}(I_i)$. During the optimization phase, the network parameters should be adjusted so that the response of these feature maps is high in locations of the image where fruits are present. To this aim, we design the following loss:

$$\mathcal{L}_j^{PAC} = \frac{1}{M} \sum_{i=1}^M \sum_{n \in S} (BCE(\sigma(\gamma(F_n(I_i))), l_i)) \quad (1)$$

$$BCE(\sigma(\gamma(F_n(I_i))), l_i) = -[l_i \log(\sigma(\gamma(F_n(I_i)))) + (1 - l_i) \log(1 - \sigma(\gamma(F_n(I_i))))] \quad (2)$$

where $S = \{1/8, 1/4, 1/2, 1/1\}$ and $\gamma(\cdot)$ is the pooling function introduced in Durand et al. (2017). $\gamma(\cdot)$ processes the input through two steps: first, a class-wise pooling operation combines the 8 response maps at the four scales, then the spatial pooling selects the relevant regions within the maps. The resulting score is compressed by the sigmoid function $\sigma(\cdot)$ and used to compute the Binary Cross Entropy (BCE) loss.

The key idea behind this objective function is the following: we need WS-FLNet to output activation maps with high responses in pixel locations corresponding to regions in the input image containing fruits. However, as already mentioned, we have neither bounding boxes nor segmentation maps that we can use to impose a pixel-wise loss, instead, only binary presence/absence labels are given. Intuitively, by observing images with and without objects, the network has to learn which elements differentiate between the two cases, and, in our application, these elements are the fruits. In practice, this implies that the back-propagated gradient, in order to minimize the BCE, has to adjust the convolutional layer weights so that the pooling function $\gamma(\cdot)$ computes a high or low score when the supervision label is 0 or 1, respectively. Since $\gamma(\cdot)$ processes the fruit maps (i.e., $F_{1/8}(I_i)$, $F_{1/4}(I_i)$, $F_{1/2}(I_i)$ and $F_{1/1}(I_i)$), the direct consequence is that the network learns to increase their responses in correspondence of fruits. The role of the four different map resolutions is twofold: first, it allows the model to learn to identify fruits regardless of their scale; secondly, it enforces consistency of presence-absence estimates across different scales. At test time, we

use only the result of the last up-convolutional stage for the fruit localization, i.e., $F_{1/1}$. Fruits are represented by the pixel coordinates of the local maxima in $F_{1/1}$. In particular, we sum up the 8 channels of $F_{1/1}$ and determine the set $U = \{(u_1, v_1), (u_2, v_2), \dots, (u_D, v_D)\}$ of pixels corresponding to the D detected fruits as follows:

$$R = \sum_{c=1}^8 F_{1/1,c} \quad (3)$$

$$U = \{(u_o, w_o) : \exists B((u_o, w_o), r) \in R \wedge R(u, w) < R(u_o, w_o) \quad \forall (u, w) \in B((u_o, w_o), r)\} \quad (4)$$

where $B((u_o, w_o), r)$ is a ball with center (u_o, w_o) and radius r . Outliers can be avoided by tuning the parameter r and by considering only the elements of U whose response in $F_{1/1}$ is greater of a fixed threshold (defined as th_U).

3.1.3. Fruit tracking

To handle occlusions and avoid redundant counts in sequential images, we track fruits across frames by using a Kalman-based multiple objects tracker. The tracking procedure is outlined in Fig. 5.

As mentioned in the previous section, a fruit is represented by the pixel coordinates of the local maxima in $F_{1/1}$ and, ideally, they correspond to its center in the image plane. We start by initializing a Kalman filter for each element in the set U of detected fruits in the first image of the stream. We define an initial state covariance matrix $\mathbf{P}_{i,1}$ and a 4×1 state vector $\mathbf{x}_{i,1}$ for the i_{th} detection as:

$$\mathbf{x}_{i,1} = [u_{i,1}, w_{i,1}, \dot{u}_{i,1}, \dot{w}_{i,1}]^T \quad (5)$$

where $u_{i,1}$ and $w_{i,1}$ are the coordinates of the fruit in the image estimated by WS-FLNet and $\dot{u}_{i,1}$ and $\dot{w}_{i,1}$ are the velocities, which we initialize to zero. The set of the Kalman filters associated to the N fruits tracked up to time step k is defined as $X_k = \{(\mathbf{x}_{1,k}, \mathbf{P}_{1,k}), \dots, (\mathbf{x}_{N,k}, \mathbf{P}_{N,k})\}$. It should be noticed that it is not practical to rely on a single Kalman filter with a state vector that includes all the tracked fruits since their number varies over time.

The Kalman prediction step for time step $k+1$ is defined as follows:

$$\begin{aligned} \tilde{\mathbf{x}}_{i,k+1} &= \mathbf{A} \mathbf{x}_{i,k} \\ \tilde{\mathbf{P}}_{i,k+1} &= \mathbf{A} \mathbf{P}_{i,k} \mathbf{A}^T + \mathbf{Q} \end{aligned} \quad (6)$$

where $\tilde{\mathbf{x}}_{i,k+1}$ is the *a priori* state estimate, $\tilde{\mathbf{P}}_{i,k+1}$ is the *a priori* state covariance, $\mathbf{Q}$ is the covariance of the process noise and $\mathbf{A}$ is the state-transition matrix, defined as:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (7)$$

Table 1

The Table reports the details of the layers that compose WS-FLNet, from the first convolution at the beginning of the network (top row of the Table) to the spatial pooling operation the computes the output (last row of the Table).

Network architecture				
Type	Filters	Size	Stride	Padding
Convolution (With BN and ReLU)	64	7×7	2	3
Max pooling	64	3×3	2	1
Residual $\times 3$	256	1×1 3×3 1×1	2 2 2	1 1 1
Max pooling	256	3×3	2	1
Residual $\times 4$	512	1×1 3×3 1×1	2 2 2	1 1 1
Residual $\times 6$	1024	1×1 3×3 1×1	2 2 2	1 1 1
Residual $\times 3$	2048	1×1 3×3 1×1	2 2 2	1 1 1
Up-convolution	512	2×2	2	1
Concatenation	2048 + 512	–	–	–
Convolution (With BN and ReLU)	512	3×3	2	1
Up-convolution	256	2×2	2	1
Concatenation	1024 + 256	–	–	–
Convolution (With BN and ReLU)	256	3×3	2	1
Up-convolution	128	2×2	2	1
Concatenation	512 + 128	–	–	–
Convolution (With BN and ReLU)	128	3×3	2	1
Convolution	8	1×1	2	1
Class-wise and spatial pooling	–	–	–	–
Up-convolution	64	2×2	2	1
Concatenation	256 + 64	–	–	–
Convolution (With BN and ReLU)	64	3×3	2	1
Convolution	8	1×1	2	1
Class-wise and spatial pooling	–	–	–	–
Up-convolution	32	2×2	2	1
Concatenation	128 + 32	–	–	–
Convolution (With BN and ReLU)	32	3×3	2	1
Convolution	8	1×1	2	1
Class-wise and spatial pooling	–	–	–	–
Up-convolution	16	2×2	2	1
Concatenation	64 + 16	–	–	–
Convolution (With BN and ReLU)	16	3×3	2	1
Convolution	8	1×1	2	1
Class-wise and spatial pooling	–	–	–	–

where Δt is the sampling period. To perform the measurement update we process the image I_{k+1} with WS-FLNet to compute $u_{i,k+1}^m, w_{i,k+1}^m, \forall i = 1, 2, \dots, N$. The fruit velocities $\dot{u}_{i,k+1}^m, \dot{w}_{i,k+1}^m$ are measured by estimating the optical flow OF between I_k and I_{k+1} . To compute the OF we employ the FlowNet2 CNN architecture and use the publicly available weights

optimized on MPI-Sintel Dataset, as defined in Ilg et al. (2017). As mentioned before, we do not perform any fine-tuning procedure, thus, we do not need to gather any optical flow annotations.

We can therefore define the set of the L measurements as $Z_{k+1} = \{z_{1,k+1}, \dots, z_{L,k+1}\}$, where the measurement vector $z_{l,k+1}$ is defined as:

$$z_{l,k+1} = [u_{l,k+1}^m, w_{l,k+1}^m, \dot{u}_{l,k+1}^m, \dot{w}_{l,k+1}^m]^T \quad \forall l = 1, \dots, L \quad (8)$$

To find the correspondence between the fruit locations estimated after the prediction step of the Kalman filter and the measurements Z_{k+1} we use a data association procedure exploiting the hungarian algorithm (Kuhn, 1955). In particular, for each candidate correspondence between the measurement $z_{l,k+1}$ and the *a priori* state estimate $\tilde{x}_{i,k+1}$, we can define a cost term as the euclidean distance between the two vectors:

$$c_{i,l,k+1} = \|\tilde{x}_{i,k+1} - z_{l,k+1}\|. \quad (9)$$

In this way, for a given assignment order m between the elements of $\tilde{X}_{k+1}$ and Z_{k+1} , we can sum all the cost terms, obtaining the total cost $C_{m,k+1}$. The best assignment is then computed by solving the following cost optimization problem:

$$\hat{C}_{k+1} = \min_{m=1, \dots, H} C_{m,k+1}, \quad (10)$$

Where H is the number of all the possible assignment combinations. New fruits that are not yet tracked are detected by setting a threshold on the cost terms $c_{i,l,k+1}$ (defined as th_c). A term in Z_{k+1} that is not associated after $T_{Z_{fruits}}$ consecutive frames (i.e., its association cost violates the threshold) is considered to be related to a new fruit previously unobserved and a new Kalman filter instance is initialized. Similarly, tracked fruits that are no longer observed are identified by checking elements in $\tilde{X}_{k+1}$ that cannot be associated with measurements Z_{k+1} in $T_{\tilde{X}_{fruits}}$ iterations. In these cases, the relative Kalman filters are deleted.

After the data association procedure, the Kalman correction stage is performed only on the elements of $\tilde{X}_{k+1}$ that have been associated with measurements in Z_{k+1} :

$$\begin{aligned} \mathbf{K} &= \tilde{\mathbf{P}}_{i,k+1} \mathbf{D}^T (\mathbf{W} + \mathbf{D} \tilde{\mathbf{P}}_{i,k+1} \mathbf{D}^T)^{-1} \\ \mathbf{x}_{i,k+1} &= \tilde{\mathbf{x}}_{i,k+1} + \mathbf{K}(\hat{z}_{i,k+1} - \mathbf{D} \tilde{\mathbf{x}}_{i,k+1}) \\ \mathbf{P}_{i,k+1} &= (\mathbf{I} - \mathbf{K} \mathbf{D}) \tilde{\mathbf{P}}_{i,k+1} \end{aligned} \quad (11)$$

where $\mathbf{x}_{i,k+1}$ and $\mathbf{P}_{i,k+1}$ are the updated state estimate and state covariance, $\mathbf{K}$ is the Kalman gain, $\mathbf{W}$ is the covariance of the observation noise, $\mathbf{I}$ is the identity matrix and $\mathbf{D}$ is the observation matrix (set to identity). The set of the updated state vectors $X_{k+1} = \{(\mathbf{x}_{1,k+1}, \mathbf{P}_{1,k+1}), \dots, (\mathbf{x}_{N,k+1}, \mathbf{P}_{N,k+1})\}$ is used for the following filtering iteration and for the next data processing phases.

3.2. Canopy/trunk detection and tracking

The detection of the canopies is necessary to associate fruits to trees, while trunk detection is used to distinguish foreground and background fruits, as discussed in the next section. As we mentioned before, the canopy/trunk detector is the sole part of the system that we decided to train with fully supervised labels. We choose this approach for two main reasons: (i) the effort needed to acquire bounding box labels to train the canopy/trunk detector is negligible compared to that one required by a fully supervised fruit detector; (ii) the same canopy/trunk detector can be used in different contexts and orchards without re-training or fine-tuning, since trees of different fruit species share similar shapes and characteristics.

The model used for the detection of canopies and trunks is based on the YOLO-v3 architecture presented in Redmon and Farhadi (2018). During the training phase, we use the weights trained on the COCO dataset, as in Redmon and Farhadi (2018), and fine-tune them on the images gathered in the orchards annotated with bounding boxes indicating canopies and trunks. The detection candidates computed by

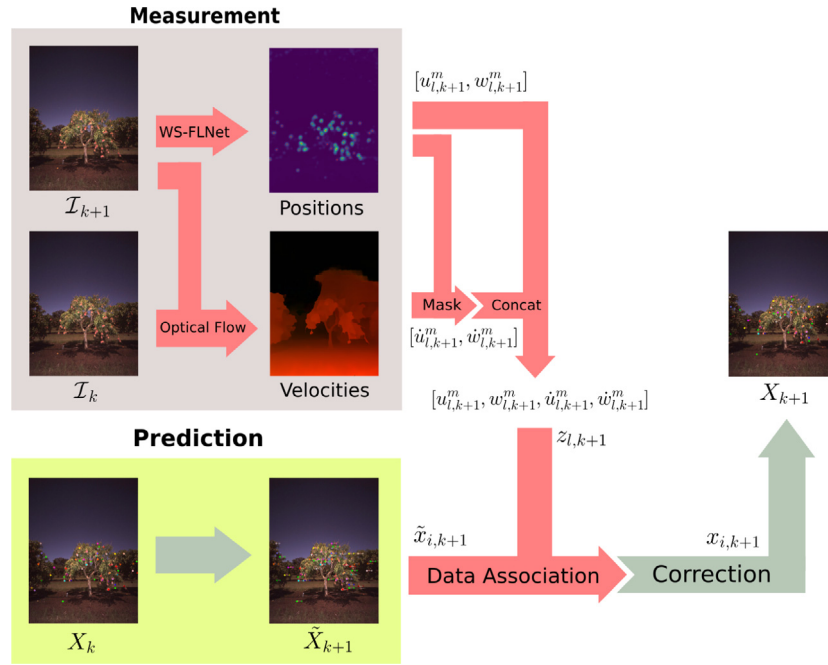


Fig. 5. The Kalman filter-based fruit tracking procedure. The Kalman prediction stage is used to propagate the *a priori* state estimates and covariance matrices. The detections estimated with WS-FLNet and the optical flow computed with FlowNet2 (Ilg et al., 2017) are used to measure the locations and velocities of the fruits. Measurements are then associated with the fruits tracked up to the previous timestep with the Hungarian algorithm. Finally, we apply the Kalman update stage on the tracked fruits that have been associated with the relative measurements.

the network are tracked across consecutive frames by using a Kalman-based multi-object tracker. Prediction, measurement update and data association procedures follow those adopted for the fruits and detailed in Section 3.1.3, except for the use of the optical flow which, in this case, is not necessary due to the low number of objects tracked.

Fruits are associated with a specific tree by simply checking whether they are contained in the associated canopy bounding box estimated by the detector.

3.3. Depth-estimation and 3D fruit positioning

To compute the 3D coordinates of the detected fruits we benefit from a depth estimation network. Fruit positioning is crucial to obtain a detailed estimation of the yield, particularly if we need to gain insights about the distribution of the fruits on different portions of the orchard. Furthermore, it allows distinguishing between foreground and background fruits with respect to the tree trunk. In this way, we can exclude from the count background fruits, which are more affected by occlusions and harder to track, and try to detect them from a better point of view at a later stage of the exploration (or acquired when navigating on the other side of the cultivation row). Also, the estimation of the detected fruit depth and the foreground/background distinction is crucial to avoid redundant counts from multiple points of view.

To perform depth estimation we rely on the DiverseDepth CNN architecture described in Yin et al. (2020). As for FlowNet2 (Ilg et al., 2017), we use the publicly available weights trained on the DiverseDepth dataset (Yin et al., 2020), as detailed in the original paper. This dataset is specifically designed to train models with better generalization capabilities to unseen scenarios. Therefore, we do not need to perform any fine-tuning procedure for our task, avoiding the need to collect depth annotated images.

3.4. Visual odometry and graph optimization

The estimation of the depth associated with each detected fruit, makes it possible to compute their 3D position with respect to the camera reference system. However, in order to perform yield estimation,

we need to build a global map containing all the detected fruits as the orchard is explored. To this aim, in this work, we choose to represent camera pose and 3D fruit positions within a graph structure.

To refine the camera poses and the fruit positions, we periodically (every k_G steps) perform a graph optimization procedure. In our case, vertices of the graph represent camera poses and 3D fruit positions, while edges encode transformations between them.

At each time step, a new vertex $\mathbf{v}_k^w$ referring to the camera pose $\mathbf{T}_k^w \in SE(3)$ (with respect to the global frame O_w) is added. This pose is computed by processing the image I_k with the VO algorithm. At the same time, an edge $\mathbf{e}_{k-1}^{k,r}$ connecting the vertices $\mathbf{v}_{k-1}^r$ and $\mathbf{v}_k^r$ is created by computing the relative increment $\mathbf{T}_k^{k-1} = (\mathbf{T}_{k-1}^w)^{-1} \mathbf{T}_k^w$ and associating the corresponding confidence matrix $\Omega_{k-1,k}^r$. The N fruits tracked up to timestep k are first projected into the 3D camera frame O_c computing $\mathbf{p}_{n,k}^c = [x_{n,k}^c, y_{n,k}^c, z_{n,k}^c, 1]$, with $n = 1, 2, \dots, N$. This is achieved by using the estimated depth map from the DiverseDepth CNN (see Section 3.3) and their 2D pixel coordinates $[u_{n,k}, v_{n,k}]$. During this process, the tree trunk and the fruit depths are compared to exclude those in the background. Afterward, we compute their 3D positions on the global reference frame O_w as $\mathbf{p}_{n,k}^w = \mathbf{T}_k^w \mathbf{p}_{n,k}^c$ and check whether they are close (up to a threshold th_G) to existing fruits in the map or not by computing the 3D euclidean distance:

1. The former case implies that the fruit has already been observed in a previous timestep (either from a different point of view or due to a loop closure) and, thus, there is already an existing vertex $\mathbf{v}_l^f$ representing it in the graph. Therefore, we do not add another vertex and, instead, create an edge $\mathbf{e}_k^{l,f}$ with relative transformation $\mathbf{p}_{n,k}^c$ and confidence matrix $\Omega_{n,k}^f$ to link $\mathbf{v}_l^f$ with $\mathbf{v}_k^r$.
2. In the latter case, the fruit has never been added in the 3D map. Hence, we create a new vertex $\mathbf{v}_n^f$ with position $\mathbf{p}_{n,k}^w$ and link it to $\mathbf{v}_k^r$ using an edge $\mathbf{e}_k^{n,r}$ associated with the measurement $\mathbf{p}_{n,k}^c$ and confidence $\Omega_{n,k}^f$.

Note that, since we are interested in the total yield of the orchard and not to that of a particular plant, we do not require our approach to link

fruits to the associated tree. Nonetheless, embedding this functionality would be trivial as we already detect trees and estimate their trunk depth. Specifically, we could simply compute the 3D positions of the trunks and use them to determine whether the tree observed in the current image is a new one or another previously mapped. In either case, the detected fruits can be assigned to that plant, allowing for tree-level production estimation.

We can now define the non-optimized graph $\hat{G}_k$ at time step k :

$$\begin{aligned}\hat{G}_k &= \{V_k, E_k\} \\ \hat{V}_k &= \{\mathbf{v}_1^r, \mathbf{v}_2^r, \dots, \mathbf{v}_k^r, \mathbf{v}_1^f, \mathbf{v}_2^f, \dots, \mathbf{v}_M^f\} \\ \hat{E}_k &= \{\mathbf{e}_1^{2,r}, \mathbf{e}_2^{3,r}, \dots, \mathbf{e}_{k-1}^{k,r}, \mathbf{e}_1^{1,f}, \mathbf{e}_1^{2,f}, \dots, \mathbf{e}_M^{k,f}\}\end{aligned}\quad (12)$$

where $\hat{V}_k$ is the set of vertices that represent the camera and fruits absolute positions, $\hat{E}_k$ contains the edges linking the vertices, and M indicates the number of fruits in the graph.

The optimization step aims to minimize the following cost function:

$$\begin{aligned}L(\Delta V_k) &= \sum_{i,j} \mathbf{d}_{i,j}^T \Omega_{i,j} \mathbf{d}_{i,j} \\ \mathbf{d}_{i,j} &= \mathbf{e}_{i,j} - ((\mathbf{v}_i + \Delta \mathbf{v}_i) - (\mathbf{v}_j + \Delta \mathbf{v}_j))\end{aligned}\quad (13)$$

where the vertices $\mathbf{v}_i, \mathbf{v}_j \in \hat{V}_k$ and $(\mathbf{e}_{i,j}, \Omega_{i,j}) \in \hat{E}_k$ is the edge connecting them, while $\Delta \mathbf{v}_i, \Delta \mathbf{v}_j \in \Delta \hat{V}_k$ are the adjustment terms of the vertices. The cost element $\mathbf{d}_{i,j}$ represents the error between the measured transformation relating vertices $\mathbf{v}_i$ and $\mathbf{v}_j$, encoded by $\mathbf{e}_{i,j}$, and the predicted one. The graph $\hat{G}_k$ and the optimal set of vertices $\hat{V}_k$ can be obtained with the calculation of all the adjustments:

$$\begin{aligned}\tilde{G}_k &= \{\tilde{V}_k, \tilde{E}_k\} \\ \tilde{V}_k &= \{\mathbf{v}_i + \Delta \tilde{\mathbf{v}}_i\} \quad \forall \mathbf{v}_i \in \tilde{V}_k \quad \forall \Delta \tilde{\mathbf{v}}_i \in \Delta \tilde{V}_k \\ \Delta \tilde{V}_k &= \underset{\Delta \tilde{V}_k}{\operatorname{argmin}} L(\Delta \tilde{V}_k).\end{aligned}\quad (14)$$

The aim of this optimization procedure is to find the optimal corrections $\Delta \tilde{\mathbf{v}}_i$ that minimize the errors between all the constraints imposed by the edges and the predicted transformation between the vertices.

For the global graph map calculation, we use the general graph optimization (g2o) method proposed in Kümmerle et al. (2011). The Visual Odometry estimation can be performed by using any vision-based monocular strategy, either model-based, such as Engel et al. (2017), Forster et al. (2016), Mur-Artal and Tardós (2017), or data-driven, such as Costante and Mancini (2020), Wang et al. (2017), Costante and Ciarfuglia (2018), Ciarfuglia et al. (2014), Zhou et al. (2017). For convenience, we use the camera poses estimated by the COLMAP approach introduced in Schönberger and Frahm (2016) and Schönberger et al. (2016).

4. Experiments

In this section, we describe the experimental setup used to validate our approach and discuss the results.

4.1. Metrics

The approach is evaluated using video sequences collected by ground robots and Micro Aerial Vehicles (MAVs) navigating in orchards with different fruit species. In order to quantitatively evaluate the performance of the overall approach, the number of fruits for the image sequences have been annotated through human supervision or obtained during the harvesting operations (note that this process is requested only in the analysis phase and it is not necessary during normal operation).

Quantitative performance for the yield estimation is computed by comparing the ground-truth values with the predictions provided by the different methods. Both our approach and the SotA methods rely on data-driven techniques to optimize different modules of the yield



Fig. 6. ACFR-Mangoes dataset (Stein et al., 2016) examples. The images are collected by driving a UGV inside a mango orchard.

estimation system. Therefore, to test the generalization capabilities of the models, we select a set of trees in each orchard to be used as the test set, i.e., they are not provided during the training phase of the different modules. In particular, we consider a test set $T = \{t_1, t_2, \dots, t_N\}$ as a collection of N trees. We indicate with t_i the ground-truth fruit count for the i th tree, while with y_i referring to the prediction of the yield estimation method. Therefore, for each tree, we can compute the Error Percentage (EP (%)) as:

$$EP = 100 * \frac{t_i - y_i}{t_i} \quad (15)$$

This metric can be positive or negative, depending on whether an over-count or an under-count is experienced. In addition, we also provide three metrics that summarize the yield estimation performance on the trees selected as the test set: the RMSE (Root Mean Squared Error), the MAE (Mean Absolute Error) and the MAEP (Mean Absolution Error Percentage):

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (t_i - y_i)^2}{N}} \quad (16)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |t_i - y_i| \quad (17)$$

$$MAEP = \frac{1}{N} \sum_{i=1}^N 100 * \frac{|t_i - y_i|}{t_i} \quad (18)$$

In the following sections, we detail the datasets used in the experiments, describe the state-of-art methods used for comparison and discuss the results.

4.2. Datasets

We use two datasets for the yield estimation experiments. The first one is the ACFR-Mangoes dataset presented in Stein et al. (2016) and contains images of mango orchards collected with an Unmanned Ground Vehicle (UGV). The second one is the ISAR-Almonds dataset proposed in Bellocchio et al. (2019, 2020), and is gathered by flying a MAV over an orchard with almond trees.

4.2.1. ACFR-mangoes

The UGV used to navigate in the orchard is developed by the Australian Centre of Field Robotics and is equipped with RGB cameras, LIDAR and GPS/RTK sensors. Images are taken with a Prosilica GT3300C camera with a Kowa LM8CX lens at 5 Hz and with a resolution of 1648×1236 pixels (see Fig. 6 for some examples). The site used for data collection is a mango orchard composed of 10 rows of trees. Each tree is observed by two video sequences taken from the two opposite sides of each cultivation row. The images are characterized by good illumination conditions that are stable across the sequence. On the other hand, fruits are often occluded by foliage and tree trunks, which increases the difficulties for the tracking module.

For comparison purposes, we choose 18 trees having different sizes and fruit numbers among all the 10 rows as the test set. The ground truth number of fruits is manually associated during the harvesting



Fig. 7. ISAR-Almonds dataset (Bellocchio et al., 2019, 2020) image examples. The images are gathered by flying a DJI Phantom 4 drone over an almond orchard.

operations. We select and use the video sub-sequences associated with these trees for performance evaluation. It is important to specify that the list of 18 trees used as test set is built following Liu et al. (2019) and it is different from the one used in Stein et al. (2016).

The remaining sequences are, instead, used to train the data-driven modules. We select 1060 images among these sequences and split them into a training and a test set, with 1010 and 50 samples each. We use the bounding box labels associated with each tree depicted in the former set to train the trunk/canopy detector. The test set is, instead, used for evaluation purposes (see Section 4.6.3). The same set is also used to train the proposed WS-FLNet for fruit detection. For this purpose, we divide all the images into smaller tiles with a resolution of 256×256 pixels and annotate each tile with the presence/absence binary label. A total of 1891 tiles are used for the training, while the remaining 98 are kept for validation. To compute the overall fruit count for each tree in the test set, we sum the estimated count for the sequences acquired from the two sides of the cultivation row containing the tree.

To evaluate the performance of the tracker modules (see Section 4.6.3), we also manually labeled 2 of the 18 tree sequences (75 images) annotating fruits and canopies with unique IDs and matching them across frames.

4.2.2. ISAR-Almonds

The ISAR-Almonds dataset (Bellocchio et al., 2019, 2020) is gathered by flying a DJI Phantom 4 drone over an orchard with almond trees (see Fig. 7). In contrast to the ACFR-Mangoes dataset (Stein et al., 2016), the image sequences exhibit significant view-point variations from one frame to another due to the frequent attitude changes of the MAV. This condition increases the complexity of this dataset and has a considerable impact on the performance of the approaches we compare, as shown in Section 4.6.2.

To train WS-FLNet and the other baselines, we separately take 17 photos at 5472×3078 pixels and crop them into smaller tiles of 300×300 pixels. We select and annotate 1057 tiles for the training set and 99 tiles for the validation set.

To train the canopy/trunk detector and evaluate the yield estimation performance, we use the video sequences of the drone flights. The video sequences have a resolution of 4096×2160 pixels and a frequency of 25 frames per second. The canopy/trunk detector optimization is supervised by selecting 1080 frames among the sequences (980 of which are used for training and 100 for module evaluation (see Section 4.6.3) and annotating them with bounding box labels of canopies and trunks. Tests are performed on the video streams associated with 6 trees, gathered from two opposite sides of the cultivation row. In this case, since there are no ground-truth counts from harvesting operations, we manually annotate all the visible fruits within the videos and separately consider the two sides of each tree during the experimental evaluation.

As for the ACFR-Mangoes dataset, we annotate 70 images of one sequence assigning unique IDs to fruits and canopies and matching them across frames.

4.3. Annotation details

The exploitation of the weakly-supervised paradigm allows us to considerably reduce the labeling effort required to train the fruit detection module. In practice, this implies a significant reduction of the time

Table 2

The table reports the parameter settings used in the experiments.

Parameter	Value
r : ball radius for the local maxima research	7 pixels
th_U : fixed threshold for the local maxima research	25.0
$P_{i,1}$: initial state covariance matrix	$diag(10.0)$
Q : process noise covariance matrix	$diag(0.1)$
W : observation noise covariance matrix	$diag(1.0)$
Δt : sampling period	0.25 s
th_c : assignment cost term threshold	20 pixels
$T_{Z_{fruit}}$: measurement non-association frame limit (fruit tracking)	2 frames
$T_{X_{fruit}}$: filter non-association frame limit (fruit tracking)	2 frames
$T_{Z_{canopy}}$: measurement non-association frame limit (canopy/trunk tracking)	2 frames
$T_{X_{canopy}}$: filter non-association frame limit (canopy/trunk tracking)	3 frames
th_G : fruit 3D euclidean distance threshold	0.08 m
k_G : Graph optimization interval	10 frames

needed to annotate images, thus extending the practical applicability of vision-based solutions for yield estimation.

To analyze the time needed for the manual annotations, we separately consider the fruit and canopy/trunk detector labeling process. As we detailed before, we divide images into tiles and label each of them. The human operator time required to determine the presence/absence of fruits in a tile is at most 1–2 s since we only need to determine whether the image contains at least one fruit and no information about the fruit positions is required. Conversely, if we rely on a fully supervised fruit detector, the bounding box annotation time depends on the number of fruits in the tile and takes, on average, 1 min. As an instance, annotating all the 17 high-resolution images from the ISAR-Almonds dataset (Bellocchio et al., 2019, 2020) (which are divided into 80 tiles each) requires more than 22 h in the fully supervised case and only 45 min in our weakly-supervised setting. Thus, we are able to reduce the labeling effort by approximately 30 times, while keeping the performance comparable to those obtained with fully supervised solutions (as shown in Section 4.6).

To optimize the trunk/canopy detector, we rely on a fully-supervised strategy. However, this procedure is deemed not excessively time consuming since we only need to annotate at most 2 canopies and 2 trunks per image. In fact, this process takes 10 s per image on average, requiring at most 3 h of time to annotate 1080 frames selected from the ISARLab Almond sequences for the canopy/trunk detector training. This implies that the annotation time, in this case, is about 12% of the time requested to annotate fruits by the fully supervised approach. Furthermore, different fruit species share similar canopy and trunks shapes, and, hence, there is no need to re-train the canopy/trunk detector for each new orchard. Therefore, it is reasonable to assume that the effect of the annotation required by the trunk/canopy detector is negligible compared to that required by the fruit detection module.

4.4. Implementation details

The different CNN architectures used in our system are implemented by using the Pytorch framework. Training is performed only for WS-FLNet and the YOLO-based canopy/trunk detector, while for FlowNet and DiverseNet we use the publicly available weights with no fine-tuning procedures.

For WS-FLNet we use a Stochastic gradient descent (SGD) optimizer, setting the learning rate to 0.001 and the batch size to 32 images. Training is performed for 100 epochs and takes 5 h for each dataset, on average. Similarly, the canopy/trunk detector, based on the YOLO-v3 architecture, is optimized with SGD by setting the learning rate and the batch size to 0.01 and 16, respectively. In this case, the learning phase lasts 150 epochs and takes 8 h.

For all the other hyper-parameters described in Section 3, the settings used in the experiments are listed in Table 2.

Table 3

Experimental results on the 18 test trees of the ACFR-Mangoes dataset (Stein et al., 2016). For each tree, we provide the number of fruits estimated by the various approaches and the error percentage with respect to the ground truth value (shown in the GT column). The + and the - signs before the error percentage refer to the over-count and the under-count cases, respectively. In addition, we also compute the MAPE, the MAE and the RMSE that summarize the performance of each approach.

Tree ID	GT	DVS (Stein et al., 2016)		MVS (Stein et al., 2016)		Liu et al. (2019)		Our	
		Estimated count	EP (%)	Estimated count	EP (%)	Estimated count	EP (%)	Estimated count	EP (%)
1	98	24	-75.51%	63	-35.71%	46	-53.06%	32	-67.34%
2	216	106	-50.92%	198	-8.33%	188	-12.96%	99	-54.16%
3	146	80	-45.20%	132	-9.58%	169	+15.75%	162	+10.95%
4	138	83	-39.85%	158	+14.49%	164	+18.84%	104	-24.63%
5	194	82	-57.73%	163	-15.97%	163	-15.97%	143	-26.28%
6	178	99	-44.38%	197	+10.67%	184	+3.37%	86	-51.68%
7	206	88	-57.28%	187	-9.22%	151	-26.69%	115	-44.17%
8	84	49	-41.66%	98	+16.66%	82	-2.38%	46	-45.23%
9	261	140	-46.36%	299	+14.55%	203	-22.22%	174	-33.33%
10	321	104	-67.60%	274	-14.64%	239	-25.54%	171	-46.72%
11	197	93	-52.79%	217	+10.15%	174	-11.67%	158	-19.79%
12	86	38	-55.81%	92	+6.97%	102	+18.60%	49	-43.02%
13	277	122	-55.95%	257	-7.22%	247	-10.83%	245	-11.55%
14	154	79	-48.70%	163	+5.84%	128	-16.88%	75	-51.29%
15	186	98	-47.31%	169	-9.13%	178	-4.30%	132	-29.03%
16	137	74	-45.98%	150	+9.48%	128	-6.56%	191	+39.41%
17	183	106	-42.07%	178	-2.73%	162	-11.47%	165	-9.83%
18	85	48	-43.52%	76	-10.58%	83	-2.35%	85	0.00%
MAEP	-		51.03%		11.77%		15.52%		33.80%
MAE	-		90.77		19.66		27.66		58.61
RMSE	-		101.11		22.98		35.16		69.79

During the test phase, the overall pipeline takes 10 seconds, on average, to process a single image. We believe that this is acceptable since yield estimation is not a critical procedure with strict real-time requirements and its computation can be partially or completely demanded to external units with high computational performance.

In our experiments, both the training and the test phase are performed using a workstation with an i7 processor, 32 GB of RAM, Ubuntu 18.04 operative system and an Nvidia Titan X GPU, with 12 GB of VRAM.

4.5. Baselines

To highlight the advantages introduced by the proposed approach, we compare it against different baselines.

On the ACFR-Mangoes dataset (Stein et al., 2016), the comparison is performed against the fully-supervised method proposed in Liu et al. (2019). Specifically, this approach relies on a vision-based monocular yield estimation system that is based on a FasterRCNN object detector (Ren et al., 2015) coupled with a semantic structure-from-motion module.

We also perform comparisons against the two versions of the method introduced by Stein et al. (2016). This yield estimation system combines a FasterRCNN object detector (Ren et al., 2015) with a LIDAR sensor to perform fruit detection and localization, while the camera pose is estimated by using a GPS/RTK system. The first version, namely the Multi-View-Suite (MVS), uses all the images of the video sequences associated with the target tree. The second version, *i.e.*, Dual View Suite (DVS), uses only the two images taken from the two sides of the cultivation row containing the target tree.

Since the code implementations of the previous baselines are not publicly available, in the ISAR-Almonds dataset (Bellocchio et al., 2019, 2020) we decided to compare our approach against other two baselines:

1. The fully supervised yield estimation baseline used in Bellocchio et al. (2020). At each step, it uses a canopy detector based on FasterRCNN (Ren et al., 2015) with a ResNet101 backbone (He et al., 2016) to process the image. The portions containing the detected canopies are then divided into smaller tiles of 300×300 pixels and processed by the fully-supervised fruit counting CNN described in Bargoti and Underwood (2017a) and based on FasterRCNN (Ren et al., 2015). The total fruit count is then computed by summing the estimated counts of each

tile. Observation of the same canopy tree across consecutive frames are associated with a Kalman-based multi-object tracker. The final yield for each tree is obtained by averaging the five highest counts estimated in the image stream. We refer to this baseline as YE-FasterRCNN (Bargoti and Underwood, 2017a) (Yield Estimation with Faster RCNN).

2. The weakly supervised version of the previous yield estimation baseline. The processing pipeline is similar to the previous one, except for the fruit counting module that is based on the weakly-supervised strategy introduced in Bellocchio et al. (2019). In the following, we refer to this approach as YE-WSCOUNT (Bellocchio et al., 2019) (Yield Estimation with WSCOUNT).

The FasterRCNN (Ren et al., 2015) canopy detectors of these two baselines are trained on the same split used for our canopy/trunk detector. On the other hand, the fully-supervised and weakly-supervised fruit counting networks are optimized on the set used for WS-FLNet (see Section 4.2).

4.6. Results

The following section discusses the qualitative and the quantitative results of the experiments run on the ISAR-Almonds (Bellocchio et al., 2019, 2020) and the ACFR-Mangoes (Stein et al., 2016) test sets.

4.6.1. ACFR-mangoes

The experimental results for the ACFR-Mangoes dataset (Stein et al., 2016) are reported in Table 3 and some qualitative outputs of our system are provided in Fig. 8. In terms of RMSE, MAE and MAEP, the best results are achieved by MVS. This is expected since this approach combines a fully supervised fruit detector, which is more robust to occlusions and background clutter, and a 3D localization method based on a LIDAR and a GPS-RTK system. The latter allows a precise robot positioning, crucial to avoid redundant counts and understand when new fruits are observed.

Although the performance achieved by Liu et al. (2019) (*i.e.*, the MAE, the RMSE and the MAEP are 27.66, 35.16 and 15.52%, respectively) is slightly worse than those of MVS, they are still comparable. This is impressive since the overall processing pipeline from Liu et al. (2019) is based solely on a monocular camera. The performance drop can be mostly explained by noticing that, in this case, the 3D positions of both camera and fruits are estimated with a structure-from-motion

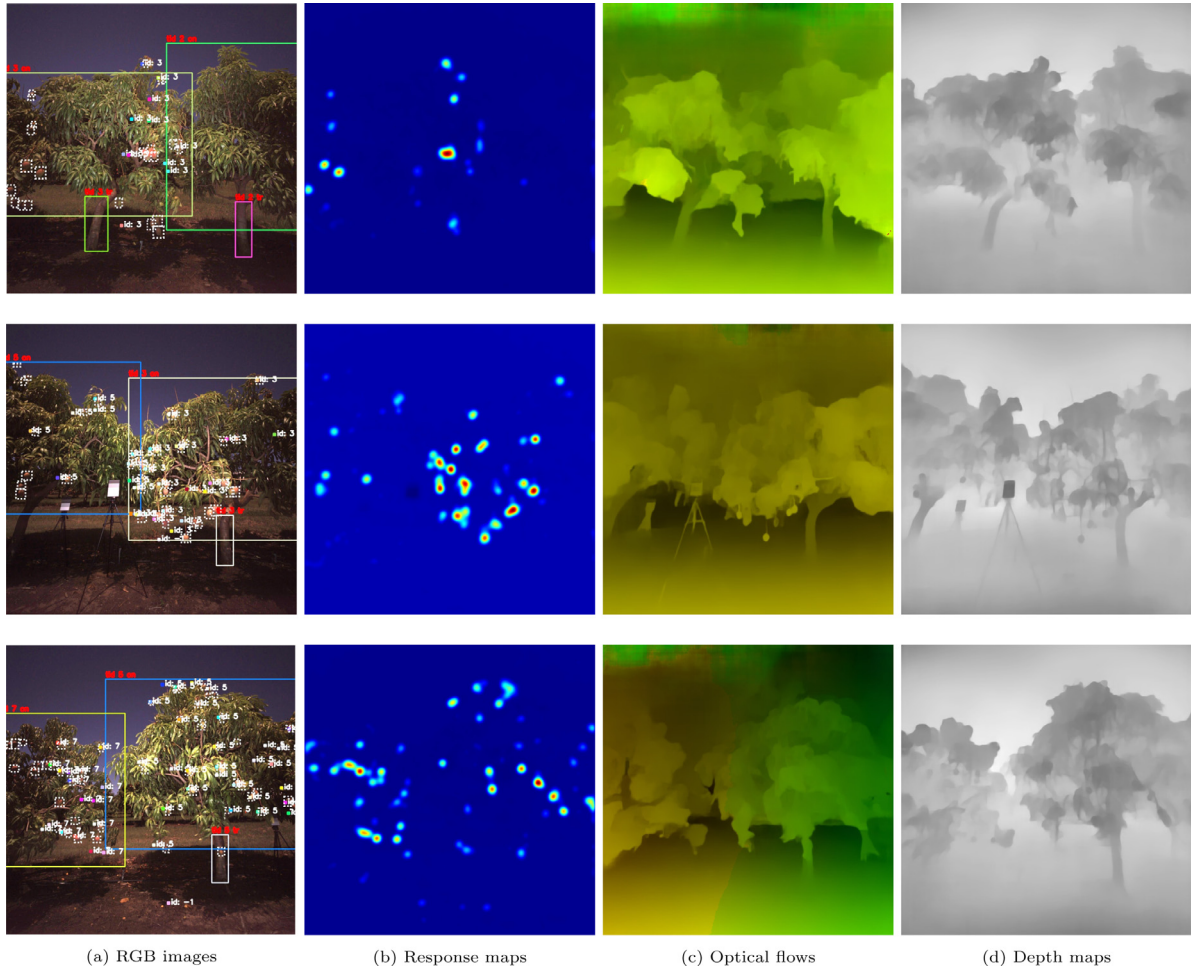


Fig. 8. Qualitative results on the ACFR-Mangoes dataset (Stein et al., 2016). From left to right, this figure shows some exemplar RGB images with fruit and canopy/trunk bounding boxes overlayed, the associated response maps computed with WS-FLNet, the optical flows estimated by FlowNet2 (Ilg et al., 2017) and the depth maps provided by DiverseDepth (Yin et al., 2020). The bounding boxes with the dotted line refer to the ground truth positions of the fruits within the images.

strategy, which is, in general, not as precise as the LIDAR/GPS-RTK system used by MVS. Nevertheless, it allows to avoid most of the redundant counts and keep the errors low. It is, however, important to notice that, as in the MVS (Stein et al., 2016) case, all the detectors are trained with fully supervised strategies.

The weak supervision paradigm we propose to train the fruit detector clearly entails a reduced accuracy if compared to the fully supervised baselines (*i.e.*, the MAE, the RMSE and the MAEP are 58.61, 69.79 and 33.80%, respectively). By analyzing the predictions for each tree, we observe that, compared to the fully supervised solutions, our method underestimates the true count, on average. This is mainly related to poor count estimates on fruits that are grouped and partially overlapped. In these situations, the fruit response maps exhibit a single cluster of pixels with high values and, hence, the local maxima search procedure will, most likely, identify only one fruit instead of multiple ones. On the other hand, fully supervised approaches are generally more effective in handling these cases since they have been explicitly trained to distinguish fruits in groups by providing bounding box labels. Nonetheless, it is important to stress that this comparison is not fair since fully supervised strategies exploit labels that specifically measure the fruit locations, while our approach relies on annotations that are significantly less informative. Our aim here is to show that, even without using highly informative labels, the performance is still satisfactory and comparable with those from MVS (Stein et al., 2016) and Liu et al. (2019). This is remarkable, considering that with our weakly-supervised solutions we are able to considerably reduce the labeling effort, as discussed in Section 4.3. In some cases, our approach

is even able to outperform the other baselines, such as for the yield estimation associated with tree 18. There are some situations, *e.g.*, tree 6, where our method performs worse. We observed that this is mainly related to the inaccurate depth map estimates from the DiverseDepth network caused most likely by the poor image conditions (*i.e.*, bad lighting conditions).

Interestingly, our approach considerably outperforms the DVS variant from Stein et al. (2016), which is fully supervised. This can be explained by noticing that this approach estimates the yield by only using the two images captured on opposite sides of the cultivation row, which are not sufficient to detect and count all the fruits within a tree. Conversely, although trained with weak supervision signals, our approach is able to provide reliable fruit detections and handles more robustly occlusions by exploiting the 3D position optimization and the fruit trackers.

It should also be noticed that all the approaches underestimate the ground truth yield for all the trees. This can be motivated by observing that, for this dataset, the actual yield is computed during the harvesting operations, when all the fruits in each tree can effectively be counted. Conversely, during image acquisition, many fruits are severely occluded by foliage or other elements and, in practice, they are never observed. Thus, it is basically impossible for yield estimation approaches to detect them.

Finally, in Fig. 9 we show the globally optimized map containing the camera and fruit locations at the end of navigation through a cultivation row. The lower part of the graph contains all the vertices expressing the camera poses, while the upper part contains the vertices

Table 4

Experimental results on the 6 test trees of the ISAR-Almonds dataset (Bellocchio et al., 2019, 2020). *side 1* and *side 2* refer to the two opposite sides of the cultivation row considered in the tests. The actual number of visible fruits for each tree is reported in the column *GT*. For each tree, we provide the number of almonds estimated by the various approaches, and the error percentage with respect to the ground truth value (shown in the *GT* column). The + and the - signs before the error percentage refer to the over-count and the under-count cases, respectively. In addition, we also compute the MAPE, the MAE and the RMSE that summarize the performance of each approach.

Tree ID	GT	YE-FasterRCNN (Bargoti and Underwood, 2017a)	YE-WSCOUNT (Bellocchio et al., 2019)	Our
		Estimated count EP (%)	Estimated count EP (%)	Estimated count EP (%)
1-side 1	88	104 +18.18%	12 -86.86%	12 -86.86%
2-side 1	140	162 +15.71%	36 -74.28%	99 -29.28%
3-side 1	85	129 +51.76%	21 -75.29%	115 -35.29%
4-side 1	36	41 +13.88%	9 -75.00%	49 -36.11%
5-side 1	42	51 +21.42%	15 -64.28%	46 +9.52%
6-side 1	275	107 -61.09%	10 -96.36%	54 -80.36%
side 1 - MAEP	-	30.34%	78.59%	46.15%
side 1 - MAE	-	44.0	93.83	64.16
side 1 - RMSE	-	71.88	124.07	97.79
1-side 2	61	75 +22.95%	9 -85.24%	9 -85.24%
2-side 2	135	153 +13.33%	10 -92.59%	7 -94.81%
3-side 2	71	92 +29.57%	6 -91.54%	19 -73.23%
4-side 2	40	76 +90.00%	13 -67.5%	16 -60.00%
5-side 2	31	51 +64.51%	2 -93.54%	7 -77.41%
6-side 2	204	129 -36.76%	5 -97.54%	7 -96.56%
side 2 - MAEP	-	42.85%	87.99%	81.21%
side 2 - MAE	-	30.66	82.83	79.5
side 2 - RMSE	-	37.15	103.05	101.45
side 1 & 2 - MAEP	-	36.59	83.29	63.68
side 1 & 2 - MAE	-	37.33	88.33	71.83
side 1 & 2 - RMSE	-	57.22	114.05	81.35

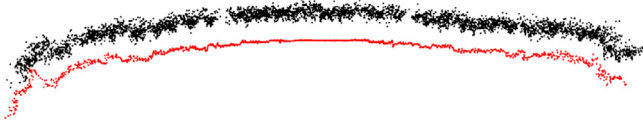


Fig. 9. In this figure, the graph map obtained from a sequence of the ACFR-Mangoes dataset (Stein et al., 2016) is shown. Black and red points represent the 3D positions of the detected fruits and the estimated camera poses, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

indicating the 3D positions of all the detected fruits. It is possible to observe that the estimated trajectory slightly deviates from the actual one (*i.e.*, a straight line), mainly due to the drifting errors of the Visual Odometry module. It is also interesting to observe that the fruits are arranged in clusters resembling the shapes of the tree canopies, confirming that our approach is able to effectively detect them.

4.6.2. ISAR-almonds

For the tests on the ISAR-Almonds dataset (Bellocchio et al., 2019, 2020), we report the results and the qualitative analysis in Table 4 and Fig. 10, respectively. When discussing the performance, in this case, we make a distinction between the results obtained on the two video sequences from the opposite sides of the cultivation rows that contain the tree in the test set. We choose to proceed in this way since, differently from the ACFR-Mangoes dataset (Stein et al., 2016), in this case, the ground truth yield is not computed during the harvesting operations. Instead, images are manually inspected to count the actual fruit number of a specific tree that can be observed from each of the two sides of the cultivation row. In this way, we are able to identify possible shortcomings or challenges related to the particular point of view from which we are observing the tree.

Differently from the more complex baselines we considered in the previous section, YE-FasterRCNN (Bargoti and Underwood, 2017a) does not explicitly handle redundant counts and occlusions by relying on 3D positioning or fruit tracking. Nevertheless, its fully supervised nature allows it to obtain the best performances in terms of MAE, RMSE and MAEP. This confirms the tendency observed in the ACFR-Mangoes

dataset (Stein et al., 2016): the fully supervised approach guarantees, on average, the lowest yield estimation errors.

On the sequence gathered from side 1 of the cultivation row, our approach is able to provide yield estimates comparable to those from YE-FasterRCNN (Bargoti and Underwood, 2017a). This is remarkable, since this dataset is considerably more challenging than the ACFR-Mangoes (Stein et al., 2016), due to UAV motion dynamics, non-ideal light conditions and varying distance from the trees. These difficulties are even more severe when side 2 of the row is observed, as shown in the last row of Fig. 10. In this latter case, we experience a performance drop compared to YE-FasterRCNN (Bargoti and Underwood, 2017a).

Nevertheless, it is more interesting to analyze the comparison against the other weakly-supervised baseline, *i.e.*, YE-WSCOUNT (Bellocchio et al., 2019). Table 4 shows that we always outperform it, particularly on the “side 1” experiments, where our approach achieves an RMSE of 97.79, a significantly lower error than 124.07 scored by YE-WSCOUNT (Bellocchio et al., 2019). This proves that the strategies we introduce to handle occlusions and redundant counts (which are not considered in YE-WSCOUNT (Bellocchio et al., 2019)) are decisive when estimating the total yield.

Finally, as in the previous section, we show in Fig. 11 the global map computed on the “side 1” sequence. In this case, the trajectory is not linear, since the drone continuously changes its height and the distance from the trees during the navigation. It is also interesting to observe that, in this case, the number of fruits varies greatly between different trees, with some of them containing very few almonds (*e.g.*, see tree 5).

4.6.3. Performance analysis of the system components

To provide a better understanding of the performance of the various components of our system, we separately evaluate the most crucial ones, *i.e.*, WS-FLNet, the fruit and canopy trackers and the canopy/trunk detectors. The first two are studied with respect to the 75 and 70 image subsets of the ACFR-Mangoes and ISAR-Almonds datasets we manually labeled as explained in Sections 4.2.1 and 4.2.2. The canopy/trunk detectors are, instead, evaluated on the test sets of the train-test splits used to optimize them (see Sections 4.2.1 and 4.2.2).

For WS-FLNet, we compute the RMSE between the number of local maxima in the response map provided by the last convolutional layer of

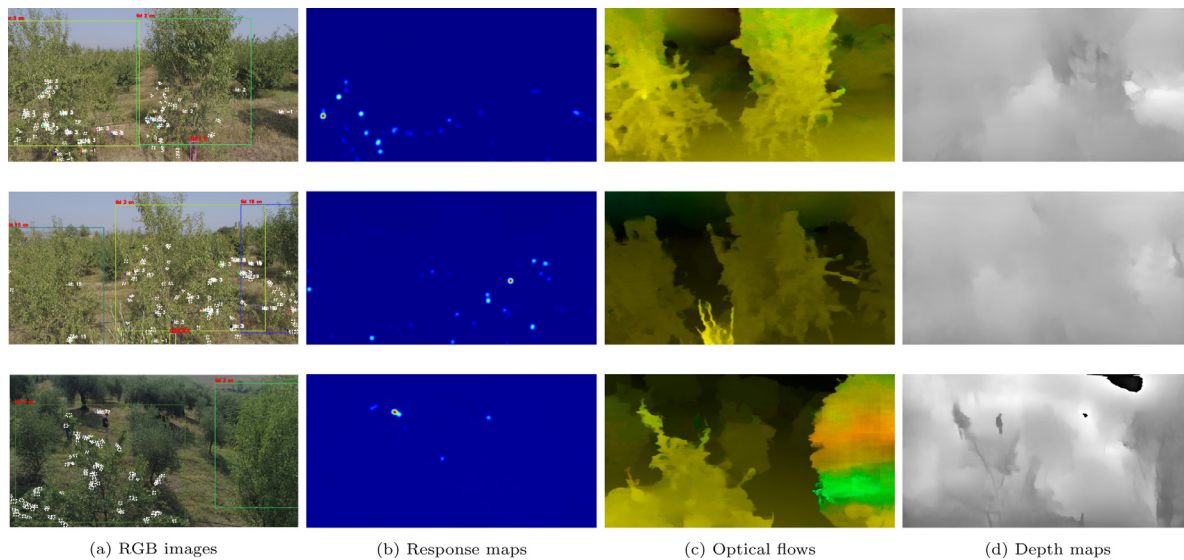


Fig. 10. Qualitative results on the ISAR-Almonds dataset (Bellocchio et al., 2019, 2020). From left to right, this figure shows some exemplar RGB images with fruit and canopy/trunk bounding boxes overlaid, the associated response maps computed with WS-FLNet, the optical flows estimated by FlowNet2 and the depth maps provided by DiverseDepth. The first two rows refer to the side 1 of the cultivation row, while the third row shows a more challenging sample from side 2. The bounding boxes with the dotted line refer the ground truth positions of the fruits within the images.

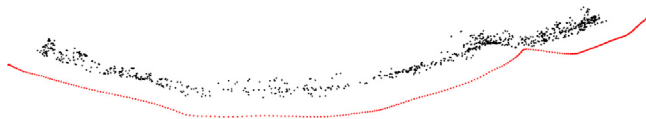


Fig. 11. This figure shows the optimized graph map obtained from the *side 1* sequence from the ISAR-Almonds dataset (Bellocchio et al., 2019, 2020). Black and red points represent the 3D positions of the detected fruits and the estimated camera poses, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the network and the ground-truth number of fruits in the image. The trackers are evaluated by using the *Multiple Object Tracking Accuracy (MOTA)* (which spans in the range $(-\infty, 1]$) (Milan et al., 2016). For the canopy tracker, we also provide the *Multiple Object Tracking Precision (MOTP)* (Milan et al., 2016), which measures the average overlap between the correctly matched objects and the ground-truth bounding boxes. Note that this metric cannot be computed for the fruit tracker, since it only provides the fruit location, without estimating its bounding box. Finally, for the canopy and trunk detectors, we use the F1 score and the mean Average Precision (mAP).

The results are reported in Tables 5 and 6. The WS-FLNet module obtains low RMSE on the ACFR-Mangoes (Stein et al., 2016) sequences, while higher errors are measured on the ISAR-Almonds (Bellocchio et al., 2019, 2020) one. This is consistent with the results of the overall system we discussed in Sections 4.6.1 and 4.6.2: the almond dataset is, indeed, more challenging due to foliage occlusions and significant view-point changes across frames. The errors of this module reflect those of the yield estimation, proving that this component is certainly one of the most crucial.

By analyzing the performance of the trackers, we can observe that the MOTA is always negative: this implies that the errors made by the tracker exceed the total number of fruits/canopies in the scene. The MOTP for the canopy tracker is between 0.201 and 0.369. This low values are expected since the estimated tree bounding boxes are heavily affected by the camera point-of-view and the foliage shape at the contours of the canopy. When analyzing the trackers' performance two aspects should, however, kept in mind. First, with respect to

other tracking scenarios (e.g., as the people ones considered in Milan et al. (2016)), fruit tracking is more challenging due to the significant presence of occlusions and background clutter, and the high similarity between different fruit instances. Second, the core of our approach is the weakly supervised detector (WS-FLNet) and it can be easily integrated with other more precise trackers, if needed.

Finally, the canopy/trunk detectors achieve good accuracies, with F1 scores above 0.80 and the mAP between 87.56% and 96.46%.

It is important to mention that we could not evaluate the depth and the 3D fruit map estimation modules since ground truth information about pixel depths and 3D fruit positions are not available for the dataset we considered.

4.6.4. Discussion on yield estimation performance

Yield estimation differs in scope and usage according to the market and the business application scenario. In particular, it is convenient to investigate which estimation errors could be acceptable for the almond and mangoes business. Historically, the production of almonds is mainly located in countries between 30° – 40° of latitude, (Reinecke, 1941), while the mango fruit is an important product in tropical regions (Rasmikayati et al., 2019). Due to the countries' diversities, it is necessary to identify an upper bound value for the error percentage of the yield estimation.

In general, underestimating and overestimating the yield does not have the same impact on the company's business. The former usually affects only the allocated resources for the production and sale and does not significantly compromise the planned market strategies. On the contrary, overestimates may expose the business to risks similar to crop loss. Therefore, the maximum value for the MAEP can be determined considering harvest loss scenarios, which mimics the production overestimation. For mangoes, typical yields fluctuations and post-harvest losses vary between the range $\pm 11 - 30\%$ (Rasmikayati et al., 2019; Savaria et al., 0000). Instead, variations with peaks of 43% were observed for almond production between 1986 and 2021 (United States Department of Agriculture, 2020). Following the above considerations, it is reasonable to set the upper-bound value for the MAEP to 30%, while it is probably acceptable to have an underestimation of up to 40%.

Table 5

Performance of the fruit and canopy trackers and the WS-FLNet module. For each sequence considered in this test, we compute the MOTA and the MOTP (Milan et al., 2016). while to evaluate WS-FLNet we measure the RMSE.

Modules		Fruit tracking		Canopy tracking		WS-FLNet
Sequence	# frames	MOTA	MOTP	MOTA	MOTP	RMSE
ACFR-Mangoes (Stein et al., 2016) sequence 1	30	−0.714	0.172	0.201	0.201	13.099
ACFR-Mangoes (Stein et al., 2016) sequence 2	45	−0.680	−0.314	0.365	0.365	37.390
ISAR-Almonds (Bellocchio et al., 2019, 2020) sequence 1	70	−0.281	−0.729	0.369	0.369	86.570

Table 6

Performance of the canopy/trunk detectors on the ACFR-Mangoes (Stein et al., 2016) and ISAR-Almonds (Bellocchio et al., 2019, 2020) datasets.

Canopy/Trunk detection module performance				
Dataset	F1 score	mAP	Precision	Recall
ACFR-Mangoes (Stein et al., 2016)	0.8173	0.9646	0.9661	0.9915
ISAR-Almonds (Bellocchio et al., 2019, 2020)	0.8109	0.8756	0.8429	0.9808

On the ACFR-Mangoes dataset (Stein et al., 2016), the fully supervised baselines, i.e., MVS (Stein et al., 2016) and Liu et al. (2019) are able to guarantee MAEP below 30%. The DVS variant of Stein et al. (2016), instead, significantly underestimates the yield (more than 50%).

Despite the performance of our strategy being inferior to some of the baselines, they are still remarkable considering that it is the sole method exploiting weak supervision. It achieves an MAEP of slightly more than 30%, close to the upper-bound value we identified. In addition, as discussed in Section 4.6.1 and shown in Table 3, in the majority of cases, errors are related to underestimations, which has less impact on the business. Considering also that, contrary to the fully supervised baselines, our approach greatly reduces the labeling burden for the training phase, we believe that it could become an important milestone for the development of efficient and fast-to-deploy yield estimation systems.

Finally, on the ISAR-Almonds dataset (Bellocchio et al., 2019, 2020), the estimation errors of the compared approaches are above the bounds we previously introduced, particularly on the *side 2* tests. As discussed in the previous sections, this scenario is more challenging and all the baselines fail to provide acceptable estimates. Nonetheless, our approach is able to outperform the other weakly supervised baseline and, on the *side 1* experiment, it is close to the underestimation bound (40%), a promising result considering the very difficult context.

5. Conclusion

In this paper, we introduced a monocular visual-based yield estimation framework, capable of tracking and mapping fruits and trees using only a single camera. We are able to significantly reduce the labeling effort needed by other fully supervised baselines by exploiting a weakly-supervised paradigm to optimize the fruit detector. Specifically, it only requires presence/absence labels during the training phase, which are considerably less burdensome to acquire than highly informative bounding boxes. As a result, the effort required to deploy vision-based solutions for yield estimation is heavily reduced, thus making possible the applications of these strategies to a wider set of real-life problems. Furthermore, we design a processing pipeline that includes tracking, depth estimation and global map optimization modules to handle redundant counts from multiple and consecutive views and occlusions.

We demonstrated through different experiments on two challenging datasets that our method is capable to achieve yield estimation performance comparable to state-of-the-art baselines that rely on fully supervised strategies (requiring expensive annotation procedures) or involve expensive sensors. In addition, our approach outperforms the weakly supervised baseline, confirming that it is an easy-to-deploy yet

reliable solution for yield estimation. Future work will focus on the robustness of the method for more challenging scenarios and fruits, including species with reduced dimensions and difficult to distinguish from the background foliage, such as olives. We will also investigate domain adaptation techniques to further reduce the need to acquire labels and try to completely eliminate the necessity to perform optimization or model tuning for each new scenario the approach is deployed into.

CRedit authorship contribution statement

Enrico Bellocchio: Conceptualization, Investigation, Data curation, Methodology, Software, Writing – original draft. **Francesco Crocetti:** Visualization, Investigation, Data curation, Resources, Software. **Gabriele Costante:** Conceptualization, Formal analysis, Methodology, Writing – original draft, Writing – review & editing, Supervision, Funding acquisition. **Mario Luca Fravolini:** Writing – review & editing, Supervision. **Paolo Valigi:** Project administration, Funding acquisition, Supervision, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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Enrico Bellocchio received the M.Sc. degree in Information and Automation Engineering From University of Perugia in 2014. He joined the Intelligent Systems Automation and Robotics Laboratory (ISARLab) in 2014 as a researcher fellow. His current research activity is focused on machine learning, computer vision and robotics.



Francesco Crocetti received his M. Sc. in information and automation engineering in 2018 from the University of Perugia, where he is currently a Ph.D. student in Industrial and Information Engineering. His research interests are mainly automation, control, robotics, and machine learning.



Gabriele Costante received the Ph.D. degree in Robotics from the Department of Engineering, University of Perugia, in 2016. He is currently a Post-Doc Researcher at the Intelligent Systems, Automation and Robotics Laboratory (ISARLab) and a Lecturer of Computer Vision at the University of Perugia, Department of Engineering. His research interests are mainly Artificial Intelligence, Robotics, Computer Vision and Machine Learning.



Mario L. Fravolini received the Ph.D. degree in Electronic Engineering from the University of Perugia in 2000. He worked as a Research Assistant in the Control Group at the School of Aerospace Engineering, Georgia Institute of Technology, and at the Department of Mechanical and Aerospace Engineering West Virginia University. Currently, he is an Associate Professor at the Department of Engineering, University of Perugia. His research interests include: Fault Diagnosis, Intelligent and Adaptive Control and Biomedical Imaging.



Paolo Valigi received the Ph.D. degree from University of Rome “Tor Vergata” in 1991. From 1990 to 1994 he worked with the Fondazione Ugo Bordoni. Since 2004 he has been Full Professor at the University of Perugia, Department of Engineering. He is currently the head of the ISARLab. His research interests are in the field of Robotics and Systems Biology.